

Desktop CPU Socket Overview

1. Intel LGA1151

Overview

The Intel LGA1151 socket became one of the company's most successful mainstream desktop platforms. It supported multiple processor generations, including Skylake (6th Gen), Kaby Lake (7th Gen), Coffee Lake (8th Gen), and Coffee Lake Refresh (9th Gen), making it a long-lasting and widely adopted standard.

Key Features

- Introduced in two electrically different revisions.
- Revision 1 supports Intel 100-series and 200-series chipsets.
- Revision 2 works with 300-series chipsets and includes additional power delivery capabilities for higher-core-count processors.
- Designed primarily for DDR4 memory, providing a solid foundation for gaming and productivity systems.
- Offered broad motherboard availability and strong ecosystem support.

Example Compatible Processor

- Intel Core i9-9900K — An iconic flagship CPU featuring 8 cores, 16 threads, and boost frequencies reaching 5.0 GHz.

2. Intel LGA2066

Overview

Intel's LGA2066 socket was developed for the High-End Desktop (HEDT) market, targeting enthusiasts, professional creators, engineers, and overclockers who required significantly more processing power than mainstream platforms could provide.

Key Features

- Paired with the X299 chipset platform.
- Supports quad-channel DDR4 memory for increased memory bandwidth.
- Provides a large number of direct PCIe lanes for graphics cards, storage devices, and expansion hardware.
- Optimized for heavy multitasking, content creation, simulation workloads, and advanced overclocking.

- Commonly found on premium motherboards with enhanced power delivery and cooling solutions.

Example Compatible Processor

- Intel Core i9-10980XE — A flagship 18-core, 36-thread processor designed for demanding rendering, encoding, and computational workloads.
-

3. AMD sTRX4

Overview

The AMD sTRX4 socket was created specifically for third-generation Ryzen Threadripper processors, delivering workstation-class performance for professionals handling highly parallel workloads.

Key Features

- Utilizes the TRX40 chipset platform.
- Offers extensive PCIe 4.0 connectivity with abundant expansion lanes.
- Supports multiple GPUs, high-speed NVMe storage arrays, and professional accelerator cards.
- Built for advanced content creation, 3D rendering, scientific computing, and enterprise-grade productivity.
- Delivers exceptional scalability for demanding workstation environments.

Example Compatible Processor

- AMD Ryzen Threadripper 3970X — A powerful workstation processor equipped with 32 cores and 64 threads, designed for intensive professional workloads and large-scale content creation.